Contractive Propagation of Perturbations as a Principle of Stability in Iterative Reconstruction and Self-Consistent Maps

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Abstract

In this work, I develop the notion of **contractive propagation of perturbations** (**CPP**) as a principle of stability for iterative problems in which noise, ill-conditioning, or lack of local contraction play a central role. In previous work, I introduced this perspective through the analysis of a transformed recurrence associated with a classical geometric method for the approximation of π. Here my purpose is different: instead of returning to the founding problem, I examine to what extent this interpretation can guide the choice of genuinely better regimes in two contemporary settings, sparse tomographic reconstruction and self-consistent maps of the *self-consistent field* (SCF) type.

I adopt a two-stage comparative methodology. First, I consider exploratory regimes and natural baselines in each problem. Then, in the most favorable branches, I deepen the analysis through parameter sweeps, multiple instances, and comparison between the best baseline found and the best ‘CPP‘ regime found. In the tomographic problem, the best ‘CPP‘ regime reduces the mean squared error relative to the best tuned baseline by gains between 39% and 56.6%, depending on the undersampling level. In the ‘SCF‘ case studied here, the improved ‘CPP‘ regime preserves robust convergence, substantially reduces the iterative cost, and outperforms the accelerated baseline tested for the map under consideration.

I do not present ‘CPP‘ as a universal algorithm. I interpret it instead as an operational principle for locating stable and productive iterative regimes in classes of problems where the propagation of perturbations is an essential part of the numerical difficulty. In that reading, the interest of the work lies both in the observable gain and in the conceptual organization that it suggests.

Keywords: iterative stability, propagation of perturbations, tomographic reconstruction, self-consistent field, inverse problems, fixed point.

1. Introduction

Iterative methods appear in many contexts in which the main difficulty is not merely to converge, but to converge without amplifying local perturbations. In inverse problems, self-consistent maps, and reconstruction schemes, small perturbations from noise, discretization, or parameter choice may grow enough to compromise the practical usefulness of the iteration, even when the problem is in principle well posed.

The question that interests me in this article is simple: when does an iteration become effectively stable? More specifically, I want to understand to what extent an interpretation centered on the propagation of perturbations can help identify better regimes within concrete iterative problems.

In previous work, I introduced this perspective through the analysis of a transformed recurrence associated with a classical geometric method for the approximation of π. In that setting, a change of variables makes explicit a particularly clean contractive structure. I do not reproduce the full development of that founding problem here. Instead, I take it as conceptual motivation

and examine what remains useful in that interpretation when one moves to broader numerical settings.

I call **contractive propagation of perturbations** (**CPP**) the regime in which small perturbations satisfy, at least locally,

$$\|\delta_{n+1}\| \leq q\,\|\delta_n\|, \qquad 0 < q < 1.$$

In that case, the factor q measures how much of the perturbation survives each step. The smaller q is, the more intense the iterative suppression of perturbations becomes.

In the founding case, ‘CPP‘ appears directly in the iterative function itself. In broader problems, however, it does not need to appear as an isolated universal operator. It may appear as a choice of variable, a choice of step size, a damped mixing strategy, or the selection of a stable iterative regime. It is in this second sense that I treat ‘CPP‘ throughout this article.

My goal is not to present ‘CPP‘ as a universal algorithm or as a general replacement for well-established techniques. The goal is more modest: to verify whether this perspective helps, in a reproducible way, locate iterative regimes that are more stable and more productive. The most consistent evidence appeared in two branches: sparse tomographic reconstruction and self-consistent maps of the ‘SCF‘ type. These are the two branches I take as the core of the paper.

2. CPP as a Principle of Stability

2.1. Conceptual origin

The initial motivation for ‘CPP‘ came from a case in which a transformed variable makes the effective dynamics of the problem explicit. The lesson I draw from that starting point is less specific than the recurrence itself: in certain situations, the numerical stability of an iteration depends decisively on the coordinate, the step size, or the regime in which the evolution is observed.

2.2. Operational interpretation

In this article, I use ‘CPP‘ in an operational sense. I look for regimes in which the update:

- does not aggressively amplify local perturbations;
- reduces sensitivity to noise or ill-conditioning;
- preserves iterative usefulness in problems where the raw form of the update is numerically fragile.

This use does not require every problem to admit a formal proof of global contraction. In many contexts, it is enough that the notion of ‘CPP‘ organize the search for a more stable regime and allow one to evaluate, through objective metrics, whether that regime actually improves numerical behavior.

3. Experimental Methodology

This section documents how I explored the usefulness of ‘CPP‘ outside the founding case.

3.1. Comparison principle

In all experiments, I followed the same logic:

1. I selected an iterative problem in which local stability and propagation of perturbations were relevant;
2. I compared one or more natural baselines with a regime guided by ‘CPP‘;
3. when one branch appeared favorable, I deepened the analysis through parameter sweeps, multiple seeds, or multiple undersampling regimes;
4. finally, I compared the best baseline found with the best ‘CPP‘ regime found.

3.2. Metrics

The metrics were chosen according to the nature of each problem:

- mean squared error (`MSE`) in tomographic reconstruction;
- convergence rate, average number of iterations to tolerance, and final fixed-point error in ‘SCF‘.

3.3. Meaning of the evidence

I interpret the computational experiments as controlled proofs of concept. The goal is not to exhaust any of the areas considered, but to verify whether the ‘CPP‘ interpretation produces observable and replicable gains in branches where iterative stability is a central part of the difficulty.

4. Sparse and Noisy Tomographic Reconstruction

The first strong branch of this work is tomographic reconstruction under undersampling and noise. The initial test was already favorable to the ‘CPP‘ regime, but I considered it important to deepen the analysis before treating it as strong evidence. For that reason, I expanded the study with multiple noise realizations, several projection levels, and an explicit sweep of the step size η.

This exploration showed that the base ‘CPP‘ regime could be improved consistently by moving to $\eta = 0.22$. After that, I also tuned the baselines in order to compare the best baseline found with the best ‘CPP‘ regime found. The result remained strongly favorable to ‘CPP‘.

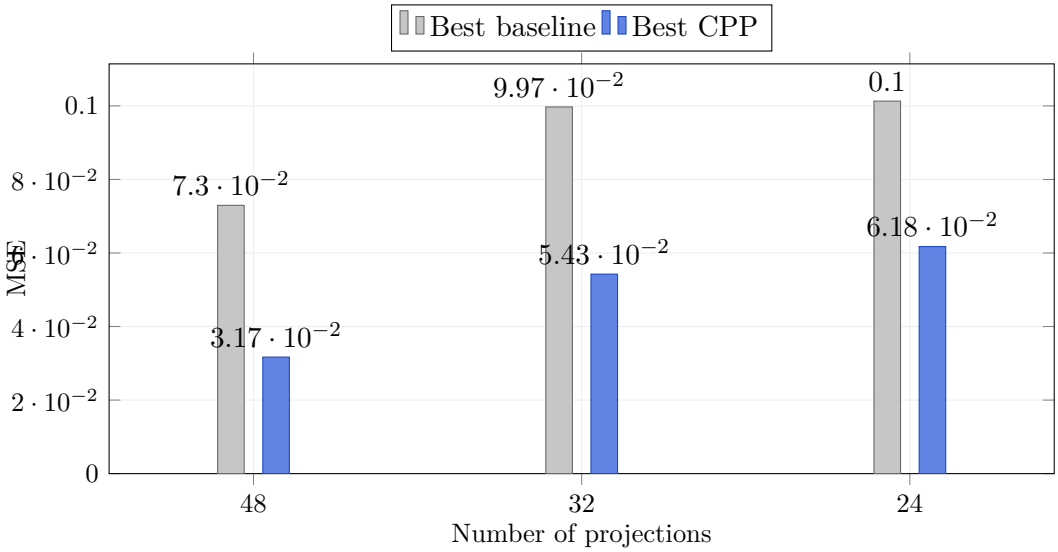

Figure 1: Comparison between the best baseline found and the best ‘CPP‘ regime in the tomographic branch across different undersampling levels.

In the final comparison, the best ‘CPP‘ regime outperformed the best tuned baseline with relative gains of approximately:

- 56.6% in MSE for the regime with 48 projections;
- 45.6% in MSE for the regime with 32 projections;
- 39.0% in MSE for the regime with 24 projections.

I interpret this result as strong practical evidence that the ‘CPP‘ perspective can help locate genuinely better reconstruction regimes in ill-conditioned inverse problems.

5. Self-Consistent Maps of SCF Type

The second strong branch provides evidence of a different kind. In the ‘SCF‘ case, the central point is not merely to improve the final quality of a reconstruction, but to stabilize the iterative dynamics itself. The deeper study showed that the raw map considered is locally non-contractive at the relevant fixed point. Even so, a damped mixing strategy guided by ‘CPP‘ produces robust convergence in all tested instances.

In this branch as well, I refined the ‘CPP‘ regime, moving from a base choice to an improved regime with $\alpha = 0.35$. The effect was a substantial reduction in the average number of iterations to tolerance, without loss of robustness.

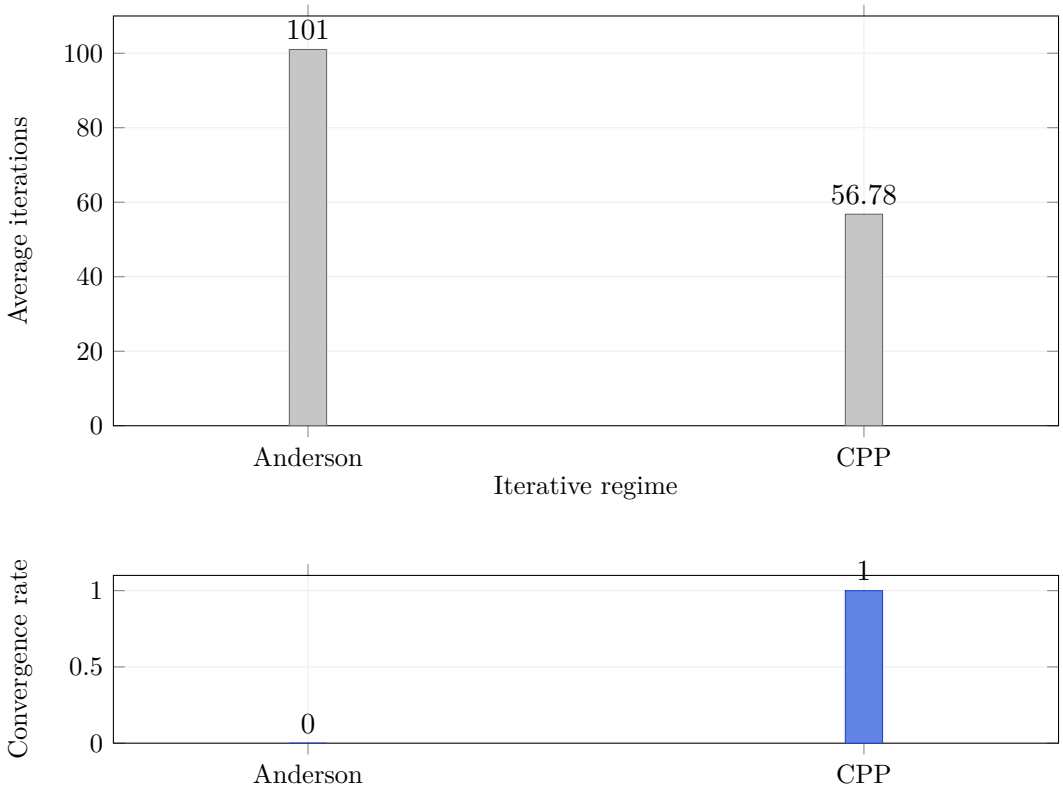

Figure 2: Comparison between the best tested baseline and the best ‘CPP‘ regime in the self-consistent ‘SCF‘ map. The upper panel shows average iterations, and the lower panel shows convergence rate.

In this map, I also compared the best ‘CPP‘ regime found with an accelerated ‘Anderson‘-type baseline. The result was unfavorable to the accelerated baseline: while the improved ‘CPP‘ regime converged in all considered instances, the tested baseline did not converge. In this branch, therefore, the main gain of ‘CPP‘ is not to accelerate an iteration that is already good, but to create the stable regime on which the iteration becomes meaningful in the first place.

6. Comparison of the Two Strongest Branches

Table 1: Confirmed gains of the best ‘CPP‘ regimes in the two most favorable branches considered here.

Area	Baseline	‘CPP‘ regime	Metric	Baseline	‘CPP‘
Tomography (48 projections)	Kaczmarz	$\eta = 0.22$	MSE	0.07296	0.03169
Tomography (32 projections)	Kaczmarz	$\eta = 0.22$	MSE	0.09972	0.05426
Tomography (24 projections)	Kaczmarz	$\eta = 0.22$	MSE	0.10130	0.06175
SCF toy	Anderson	$\alpha = 0.35$	average iterations	101.00	56.78
SCF toy	Anderson	$\alpha = 0.35$	convergence rate	0.00	1.00

The two strong branches play different roles in the argument of this work. Tomography shows direct applied value: the improved ‘CPP‘ regime produces measurable reconstruction gains against tuned baselines. ‘SCF‘ shows structural value: the ‘CPP‘ regime transforms a map that

is unfavorable to direct iteration into a robust and productive process. Taken together, these results suggest that ‘CPP‘ has greater potential when it is used as a stabilization principle in iterative problems in which noise, ill-conditioning, or lack of local contraction play a central role.

7. Scope and Limitations

I consider it important to make the limits of the proposal explicit so that the reach of the results is not overestimated.

7.1. CPP is not a universal method

The exploratory experiments carried out outside this core do not support the idea that ‘CPP‘ should, in general, replace classical algorithms or specialized methods. In several problems it was only competitive, and in others it was clearly behind approaches that were more adequate to the physical or geometric structure of the problem.

7.2. Dependence on the problem type

The most favorable results appeared when three ingredients were present at the same time:

- explicit iteration;
- relevant propagation of noise, error, or local perturbations;
- real risk of amplification due to aggressive updates or ill-conditioning.

By contrast, ‘CPP‘ tended to be less useful when a structurally specialized method for the problem already existed or when damping corrections was not the appropriate mechanism for selecting the correct solution.

7.3. Meaning of the contribution

I see the main contribution of this article less as the introduction of a universal algorithm and more as the identification of a principle of stability that, in certain contemporary iterative branches, helps locate more robust and more productive regimes. If that interpretation is correct, then the value of the work lies in showing that the propagation of perturbations can be used not only as a posterior diagnostic, but also as a criterion for numerical design.

8. Conclusion

In this article, I treated ‘CPP‘ as a principle of stability for iterative problems in which the propagation of perturbations is an essential part of the numerical difficulty. Instead of returning to the founding problem in its entirety, I used it as conceptual motivation and concentrated the analysis on two branches in which the perspective proved particularly fertile: sparse tomographic reconstruction and self-consistent maps of the ‘SCF‘ type.

In both cases, the evidence went beyond an isolated test. I carried out parametric refinement, comparison against tuned baselines, and identification of improved ‘CPP‘ regimes. In tomography,

this produced direct reconstruction gains. In ‘SCF‘, it produced clear gains in stability and efficiency. I interpret this set of results as an indication that ‘CPP‘ can function, in certain contexts, as an operational principle for locating genuinely better iterative regimes.

I do not conclude that ‘CPP‘ is a universal theory or a ready-made algorithm for every class of problem. My conclusion is more modest and, in my view, more useful: when the main difficulty of an iteration lies in how local perturbations propagate, a contraction-oriented interpretation can help transform a numerically fragile dynamics into a more favorable regime. I see in this a concrete possibility of linking conceptual stability and practical gain.

Declarations

Funding The author received no financial support for this research.

Conflicts of interest The author declares that there is no conflict of interest.

Data availability The computational experiments in this study were generated from synthetic data and controlled numerical setups created by the author. The scripts and derived summaries used in the study are available in the associated study folders.

Code availability The code used to generate the computational experiments, tables, and summaries discussed in the manuscript is made available by the author under the GNU AGPLv3 license in the project repository and in the associated study folders.

Ethics approval Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

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